Socially Embedded LLMs as Social Facilitators: Evidence from a Natural Experiment on X.com

Abstract

The integration of generative AI (Gen-AI) into social platforms is fundamentally reshaping how users discover information and interact online. This study examines an emerging class of platform native generative systems we term *Socially Embedded LLMs* (SELs), which draw on real-time user content to synthesize contextual responses and route user attention back into the platform's social graph. We analyze the release of Grok on X.com, introduced to non-premium users in New Zealand in November 2024, but not elsewhere. Exploiting this release, we employ a difference-in-differences (DID) design using Australia as the control group. Results show that Grok increases engagement with others' posts (favorites, replies, and reposts) by lowering discovery barriers and providing interaction-ready entry points to real-time content. These effects are most pronounced for low-popularity topics, suggesting that SELs can revitalize the "long-tail" of discussions. Mechanism analyses further suggest that Grok enhances engagement, in part, by increasing participant and topic diversity, thereby attracting broader audiences and fostering conversations across topical clusters. These findings provide early causal evidence that, when designed with embedded social entry points, SELs function as facilitators rather than information substitutes, highlighting their potential to rebalance attention and strengthen the resilience of social media ecosystems.

Keywords: *Generative AI, Socially Embedded LLMs, Social media, User engagement, Natural Experiment*

1 Introduction

Digital platforms are increasingly integrating generative artificial intelligence (GenAI)[1] into their core user workflows to alter how individuals discover information. Moving beyond the initial wave of stand-alone applications such as ChatGPT, Large Language Model (LLM) based Gen-AI is now embedded within platforms to support active discovery and reduce the effort required to consume content (Kim et al. 2024, Shan and Qiu 2025). Social media platforms are at the forefront of this trend, transitioning users' active information seeking from keyword-based retrieval to dynamic conversational interaction. As this shift accelerates, a critical question emerges: *Does integrating Gen-AI into a platform's social infrastructure increase or crowd out the human-to-human interactions that drive network value?*

This question highlights a fundamental design tension between cognitive effort and social connectivity in the context of active information retrieval, as opposed to passive exposure to platform-recommended content. On the one hand, traditional social media platforms often require users to deliberately seek out relevant content themselves, typically through keyword-based search as a primary discovery mechanism. This approach imposes a significant cognitive burden, as users must individually filter and aggregate fragmented results to make sense of ongoing conversations (Pirolli and Card 1999, Jones et al. 2004). On the other hand, stand-alone LLM interfaces (e.g., ChatGPT-style assistants) can lower cognitive effort through in-situ synthesis but risk creating participation dead-ends by satisfying users' informational needs without directing them back to the community (Cao et al. 2024, Jiang et al. 2025).

This tension is further compounded by the design of existing LLM reference systems. Currently, when stand-alone LLMs cite their sources, they typically do so by appending external website URLs to their synthesized text. However, these links function primarily as verification tools rather than social entry points, and using them for community engagement imposes substantial interaction costs, as it requires users to switch contexts and navigate external interfaces. This friction may also lead users to accept the LLM's synthesized response as an endpoint, decoupling individual information consumption from the organic feedback loops that sustain a social media platform's social ecosystem. Therefore, a critical challenge for digital platforms is to deploy LLMs that reduce the cognitive load of discovery without diminishing the social interactions that drive network value.

A prominent industry response to this design challenge is X.com's (formerly Twitter) introduction of Grok.[2] Unlike stand-alone LLMs that, at most, monitor social data externally, Grok is integrated directly into

[1] The global Generative AI market was valued at USD 19.67 billion in 2024 and is projected to reach USD 324.68 billion by 2033, growing at a compound annual growth rate of 40.8% (Grand View Research 2025).

[2] There are increasing number of social platforms incorporating LLM with similar objectives. Snapchat's My AI (Snapchat 2023), LLMs implemented across Facebook, Instagram and WhatsApp (Meta 2024), and Diandian in Rednote are a few examples (Tech in Asia 2024).

the platform's active discovery workflow (see Figure 1). When prompted, it retrieves real-time posts, synthesizes a narrative summary, while embedding direct links to the original posts within X.com, allowing users to immediately click through, read, reply, or repost. Figure 1 provides an illustrative example of this interface[3]. In contrast to the informational references found in stand-alone LLMs, which often require switching to an external website to verify, Grok's embedded links serve as low-friction social entry points that route users seamlessly back into ongoing conversations. This design could bridge the gap between the AI-generated synthesis of existing content and participation in ongoing conversation, offering a concrete realization of how AI might reduce the cognitive load associated with discovery without severing the social loop observed in stand-alone LLMs.

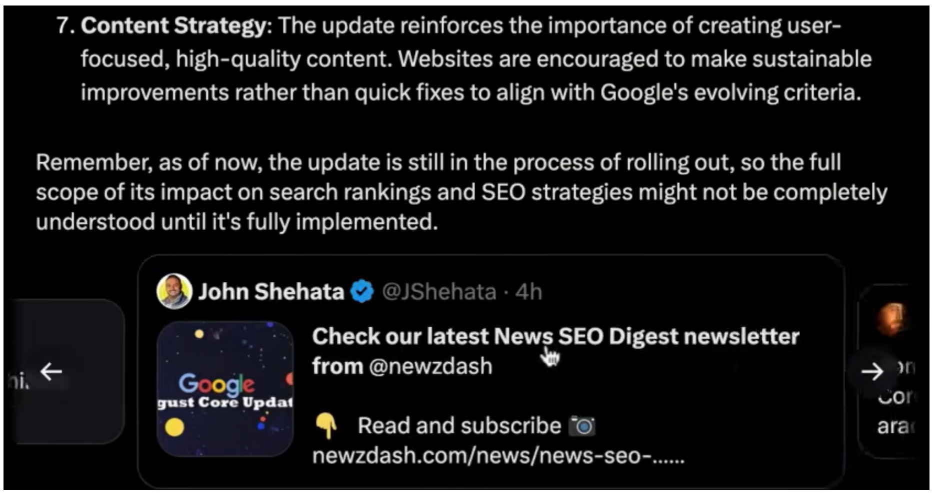

Figure 1. Grok on the X.com Platform

We conceptualize this emerging class of systems as *Socially Embedded LLMs* (SELs). We define a SEL as a platform native generative-AI system that lowers the cognitive load of information discovery while structurally routing user attention back into the interdependent social graph[4]. The defining characteristic of an SEL is not only data access (i.e., proprietary access to platform content is a necessary but insufficient condition) but rather placement within the user's action loop. This platform's native integration eliminates the barrier between passive reading and active participation. In stand-alone LLMs, moving from AI-generated content to a live discussion requires significant effort (such as the friction of switching platforms), whereas in SEL, the transition is seamless. This converts the AI from a terminal, where users stop once satisfied, into a conduit, where users are routed back to the community. In turn, it creates an interactive cycle in which the AI

[3] Figure 1 illustrates this workflow during our estimation period using the query, "What is the latest news on the recent Google core algorithm update?" In this example, Grok presents a synthesized answer (top part of Figure 1) together with integrated reference links to the underlying posts (bottom part of Figure 1) based on real-time X.com posts. Clicking these links redirects users to original social media content on X.com.

[4] Prominent examples of SELs include Grok on X.com and Diandian on Rednote (Xiaohongshu).

simultaneously draws from and actively shapes ongoing social discourse by connecting and surfacing content. This unique setting allows us to isolate the impact of integrated generative discovery brought by SELs, motivating our first research question: *What is the causal effect of SEL adoption on user engagement with other users' posts?*

Despite the theoretical promise of SELs as social conduits, prior research has largely viewed LLMs through the lens of individual augmentation. Existing studies have robustly demonstrated that stand-alone LLMs can improve performance in text-based tasks (e.g., writing and ideation) and increase efficiency in user production outcomes (Noy and Zhang 2023, Doshi and Hauser 2024, Shan and Qiu 2025). In these contexts, users often treat the LLM as a stopping point that satisfies the immediate task in situ. However, far less is known about how SEL functions as a social intermediary between a user and a community, and it remains empirically unclear whether the conduit mechanism dominates in practice.

Existing theory offers competing predictions. On the one hand, following the cognitive load theory, lowering the cognitive load of discovery can free user attention for downstream activity (Pirolli and Card 1999). Recent evidence suggests that reducing cognitive effort through integrated AI features can increase users' engagement (Kim et al. 2024). Extending this logic to social platforms, SELs like Grok function as a complement by synthesizing fragmented results within the platform, thereby freeing up cognitive resources that users can reallocate toward active participation (e.g., replying and reposting). On the other hand, cost-reducing assistance can also induce substitution: when platforms make information dramatically easier to consume, users may behave as cognitive misers, satisficing with the AI-generated summary and under-investing in broader exploration(Jiang et al. 2025). Consistent with this concern, when content is integrated in-situ, it fundamentally lowers the cost of extraction but simultaneously accelerates satiation, which can reduce downstream engagement (Cao et al. 2024). If SEL satisfies the user's curiosity in situ, the reduction in discovery friction becomes irrelevant because the motivation to click through is diminished. As such, it remains an open empirical question whether the reduction in cognitive cost is sufficient to outweigh the substitution effect driven by informational satiation.

To empirically probe whether cognitive load reduction indeed drives observed engagement gains, we examine the moderating role of topic popularity. For highly popular topics, discovery costs are already low because platforms surface them prominently (e.g., trending topics and widely adopted hashtags), making it easy for users to notice and locate relevant conversations (Carrascosa et al. 2015, Zhang and Ng 2023). In such settings, the marginal cognitive relief provided by an SEL is limited. In contrast, niche topics generally face higher cognitive barriers for discovery. By lowering search effort and surfacing overlooked content, SELs may redirect attention toward less visible discussions, potentially revitalizing long-tail participation (Brynjolfsson et al. 2010). This expectation aligns with prior evidence that AI assistance yields the greatest benefits where baseline support is weakest (Noy and Zhang 2023, Brynjolfsson et al. 2025). Our second research question therefore asks: *Does topic popularity moderate the effect of adopting SEL on user engagement?*

We also examine the mechanisms through which SELs influence user engagement. Traditional platform designs often silo users, which can limit exposure to diverse viewpoints and make discovery beyond one's established social circle more effortful (Bakshy et al. 2015, Barberá et al. 2015). We argue that SELs relax these constraints through a dual diversification mechanism. First, by synthesizing complex context, SELs lower the cognitive barrier to enter new domains, encouraging topic diversification and broadening users' topic repertoire. Second, engaging with unfamiliar users beyond one's existing social proximity imposes a significant cognitive burden, such as the effort required to locate relevant strangers and to evaluate their intent and credibility. The implementation of SELs lowers both the friction and perceived risk of this cross-cutting exposure, making it cognitively affordable to interact with a more heterogeneous mix of participants and bringing in fresh voices that inject information novelty into ongoing discussions. Consequently, we view the expansion of both the topic and participant bases as the observable downstream behavioral manifestation of reduced cognitive load, which ultimately drives higher engagement with others' posts. This motivates our third research question: *Do increases in topic and participant diversity mediate the effect of SEL adoption on user engagement?*

To address these research questions, we leverage a natural experiment created by Grok's staggered rollout on X.com. On November 10, 2024, X.com introduced Grok to non-premium users in New Zealand for the first time, while it remained unavailable to non-premium users elsewhere[5] (Mehta 2024). Using Australia as a control group, we apply a difference-in-differences (DID) approach to identify the causal effect of Grok access on user engagement.

Building on this empirical strategy, our analysis documents three key patterns. First, Grok's introduction leads to a significant rise in engagement with others' content on X.com, as reflected in favorites, replies, and reposts, suggesting that the benefit from cognitive reduction outweighs the substitution effect. Second, consistent with our heterogeneity hypothesis, the effects vary across topics: the rise in engagement is most pronounced for initially less popular topics, while highly popular topics (which are already easily discoverable) show no significant change. Given that the treatment effect is concentrated in these niche discussions, we focus our follow-up analyses on this subset to isolate the mechanism. Finally, we find that Grok adoption is associated with a broader scope of interaction, significantly increasing the entropy of both the participant base and users' topic repertoire. A subsequent mediation analysis suggests that this dual diversification serves as a key pathway through which Grok enhances engagement.

This study makes several contributions to research on Gen-AI and digital platforms. First, we explore a novel class of Gen-AI on social media platforms, SEL, and provide among the earliest causal evidence on its net impact on user-to-user engagement on social media platforms. We theorize and empirically validate the construct, distinguishing it from stand-alone LLM assistants, and demonstrate that, rather than substituting for

[5] Our empirical sample strictly excludes all premium users. Detailed procedures are discussed in Section 3.1.

organic interaction, SELs can facilitate engagement by surfacing relevant information and lowering the effort required to engage with others' posts. Second, our heterogeneous effect analysis advances understanding of platform engagement dynamics by showing that the impact of SELs is not uniform. Their value lies primarily in revitalizing the long tail (i.e., less popular topics), where they reduce entry barriers for users who might otherwise struggle to participate. Third, we identify increased diversity as a key mechanism through which SELs increase user engagement. This contributes to the scholarly conversation on platform homophily and echo chambers, providing novel evidence that AI-driven content curation can reshape the social composition of online discussions. We show that SELs are associated not only with the intensity of engagement (volume) but also with their breadth, potentially widening the participant pool and mitigating social fragmentation. The findings further offer practical guidance on how platforms can strategically deploy SELs to support user engagement while also enhancing the diversity and resilience of their content ecosystems.

2 Theoretical Foundation and Hypotheses Development

2.1 From Chatbots to Socially Embedded LLMs

Virtual assistants such as Siri and Alexa brought conversational AI into daily life (Wald et al. 2023), supported by advances in speech and affective computing that enabled emotion recognition (Lin et al. 2024) and voice-based interaction (Wang et al. 2023). Chatbots have been extensively used in customer-support settings, automating responses within defined functional boundaries (Luo et al. 2019, Jiang et al. 2022). Although these assistants expanded conversational reach, their communication style are command-driven and task-bounded rather than adaptive to evolving social dynamics (Luger and Sellen 2016).

The introduction of Large Language Models (LLMs) shifts conversational systems from scripted flows to intent-driven and reasoning-capable agents. Recent work documents productivity gains in individual and organizational contexts, such as improving professional writing (Noy and Zhang 2023) and enhancing customer service performance (Brynjolfsson et al. 2025). Evidence on platform-level effects is mixed: while LLM access can enhance efficiency, it may also impose cognitive burdens (Shan and Qiu 2025) and reduce content production (Del Rio-Chanona et al. 2024, Burtch et al. 2024). In creative domains, LLMs increase output and peer evaluations, while reducing novelty (Zhou and Lee 2024). Reliance on automated judgments may distort collective decision-making (Gao et al. 2025) and reduce diversity of creative outputs when individuals draw from similar AI-generated ideas (Doshi and Hauser 2024).

Despite these advances, most research considers the scenario in which an LLM serves as a private assistant that enhances individual performance (Luo et al. 2019, Jiang et al. 2022, Chandra et al. 2022) rather than as a facilitator of social interaction among users. This view largely overlooks the interdependent nature of user interactions on social platforms, where engagement depends on how users encounter and react to one another's content (Shi et al. 2014, Shore et al. 2018). In contrast, SELs represent a distinct platform native generative design architecture that draws on real-time user content to produce contextual responses and guide users toward participation in ongoing discussions. Our research is also related to the emerging literature on social bots, in which Gen-AI operates as an active user, either representing an influencer (Zhang et al. 2026) or a platform commenter (Gao et al. 2025). In these studies, users communicate directly with the AI, which, as Gao et al. (2025) demonstrate, can lead to a substitution effect in which engagement becomes trapped in a human-bot loop rather than fostering human-human connections. However, in our setting, SEL is not acting as a conversational peer; rather, it functions as a facilitator and discovery tool that actively routes users' attention back into the platform's human-to-human social graph.

Importantly, SELs differ from existing active information retrieval methods in two key ways. First, unlike stand-alone LLMs (e.g., ChatGPT), which provide self-contained answers that often substitute for community interaction (Lin et al. 2024, Shan and Qiu 2025), SELs operate as triadic intermediaries embedded in the platform's social infrastructure. By placing interaction-ready links directly into their responses, they function not as information dead ends but as gateways that route attention back to human users. Second, SELs fundamentally differ from traditional search, which relies on the user to manually go through and synthesize fragmented results. While search functions as an active retrieval tool, it imposes a high cognitive cost, requiring significant effort to reconstruct a narrative from disparate posts. SELs retain the active nature of information retrieval but substantially reduce this cognitive load by automating the synthesis process. This reframes the AI from a productivity tool into a social intermediary of attention and invites inquiry into how embedded agents reconfigure the interdependent dynamics of digital ecosystems.

2.2 The Impact of Socially Embedded LLMs on User Engagement

Recent studies have begun to explore the social and behavioral implications of Gen-AI, which presents a tension between two streams of research. One stream of research suggests that Gen-AI may substitute for community interaction by encouraging low-effort information processing. This possibility aligns with the cognitive miser view that individuals conserve cognitive resources by relying on simplifying shortcuts rather than engaging in effortful reasoning, especially when doing so reduces costs (Stanovich 2009, Jiang et al. 2025). In the context of Gen-AI, this tendency implies a risk of shallow processing: users may rely on convenient AI outputs instead of engaging deeply in the underlying discussion. Faced with complex and information-rich environments, individuals often rely more on fast, intuitive judgments than on deliberate reasoning (Stanovich 2009, Moritz et al. 2014). When Gen-AI produces coherent, self-contained responses to a user prompt, it can reduce search and processing effort, creating a "low-friction path" that appeals to this instinct (Branco et al. 2016, Risko and Gilbert 2016). As a result, users may substitute away from effortful exploration of the original discussion toward consuming the AI-generated output when reduced costs trigger cognitive-miser behavior (Jiang et al. 2025).

Importantly, while stand-alone LLMs can append external links to their generated content, these citations do little to change the underlying dynamics. Because they primarily function as static verification tools rather than interactive social conduits, using them requires users to navigate to external websites, imposing a significant cognitive and interaction cost of context switching.

Empirical evidence from knowledge-sharing communities (e.g., Stack Overflow) also suggests that stand-alone LLMs can substitute for human interaction. By providing satisfactory and self-contained answers, stand-alone LLMs can improve individual productivity but simultaneously reduce incentives for public contribution and community participation in knowledge-sharing communities (Burtch et al. 2024, Sanatizadeh et al. 2025, Shan and Qiu 2025). Applied to SELs, the same logic predicts that if SELs (e.g., Grok) outputs are perceived as sufficiently comprehensive, users may shift from active exploration to passive consumption, reducing engagement with others' posts.

Conversely, a second stream of research suggests that AI assistance may facilitate engagement by mitigating cognitive overload. Social content processing demands substantial cognitive resources (Sweller 1988, Bawden and Robinson 2009), and when information inflow exceeds users' processing capacity, users withdraw or discontinue (Eppler and Mengis 2004, Fu et al. 2020). Providing structured, simplified information can reduce overload and lower search and processing costs, improving users' understanding and evaluation of content (Jones et al. 2004, Jiang and Benbasat 2007, Branco et al. 2016). Applying this logic to SELs, AI-generated content can reduce cognitive load by synthesizing scattered posts into a coherent narrative, enabling users to assess relevance quickly with less effort. By lowering comprehension costs, SELs may help relocate cognitive resources toward active engagement. Similar patterns emerge in recent studies, where AI summaries have been shown to stimulate higher commenting activities on video platforms (Kim et al. 2024) and stimulate browsing in e-commerce (Su et al. 2024). A critical gap in this literature, however, is that the Gen-AI tools studied typically function as informational terminals, optimizing consumption efficiency but typically operating outside the platform's social interaction pathways. In contrast, we conceptualize SELs as social intermediaries that can route users back to the community via embedded links, converting the efficiency of synthesizing user-generated content into a direct pathway for social connection. In this sense, SELs can create sustainable value when embedded within existing social practices and organizational routines (Berente et al. 2021).

We argue that, for SELs on social platforms, facilitation is likely to dominate substitution for two reasons. First, the nature of the user objective differs fundamentally from previous settings. Unlike utilitarian platforms (e.g., Stack Overflow), where the goal is obtaining an answer to a coding question, the primary utility on X.com is social connection and self-expression. An AI-generated answer can satisfy a user's informational need, but it cannot replace the expressive need to voice an opinion or signal agreement with peers. Second, the embedded links in SEL could significantly reduce the participation cost on social media platforms. In prior studies on stand-alone LLMs, verifying the AI's answer required high-effort manual navigation. SELs mitigate this by providing interaction-ready citations. By lowering the interaction cost to near zero, the agent effectively neutralizes the cognitive barrier to entry. Consequently, SEL generated content serves not as a replacement for the social thread but as a facilitator that allows users to pursue their social goals at lower cognitive expense. Thus, we propose:

Hypothesis 1 (H1). *The introduction of SELs increases user engagement as measured by favorites, replies, and reposts.*

2.3 The Moderating Role of Topic Popularity

Social media platforms such as X.com enable information to spread far beyond direct connections through reposting and broadcast-style exposure mechanisms (Bakshy et al. 2012, Shi et al. 2014, Goel et al. 2016). In such environments, visibility and attention are central determinants of downstream engagement (Wu and Huberman 2007, Shi et al. 2014). However, attention is highly uneven: a small set of topics captures disproportionate exposure, while the majority receives little attention, producing a long-tail distribution of visibility (Merton 1968, Salganik et al. 2006, Weng et al. 2012). This imbalance is reinforced by platform mechanisms, such as trending hashtags (Yang and Peng 2022, Zhang and Ng 2023) and algorithmic curation and ranking systems that shape what content users are exposed to (Bakshy et al. 2015, Huszár et al. 2022, Guess et al. 2023).

For highly popular topics, discovery costs are already low: users can easily identify central themes and entry points for engagement through the platform's native discovery features. In such saturated environments, incorporating SELs is likely to yield only marginal incremental value over the platform's existing discovery mechanisms. In contrast, low-popularity topics face higher discovery barriers. Because collective attention is a fundamentally scarce resource, these niche discussions are disproportionately crowded out by high-visibility content (Wu and Huberman 2007, Weng et al. 2012) and receive less social amplification (Bakshy et al. 2012, Shi et al. 2014). Consequently, users must invest more effort to locate, filter, and interpret these peripheral threads (Chi et al. 2000, Pirolli 2005). We argue that SELs' discovery functions provide the greatest utility precisely in such high-cost environments. By synthesizing scattered information and offering direct entry points into otherwise hard-to-find or overlooked discussions, SELs lower the costs required for participation and facilitate engagement with long-tail topics (Pirolli and Card 1999, Chi et al. 2000, Branco et al. 2016). In this sense, the moderating effect of topic popularity could also serve as an empirical validation that the reduction of cognitive effort is an important driver of SEL-induced engagement.

This dynamic mirrors evidence on Gen-AI productivity, which shows larger marginal gains for less experienced or lower-performing users (Noy and Zhang 2023, Brynjolfsson et al. 2025). Similarly, we posit that SELs disproportionately benefit niche topics by lowering discovery and participation costs, thereby increasing engagement in the long-tail of discussion:

Hypothesis 2 (H2). *The positive effect of SELs on user engagement will be stronger for topics with low initial popularity.*

2.4 The Mediating Role of Topic and Participant Diversity

User diversity plays a central role in sustaining engagement within online discussions. Prior research shows that when online information consumption is concentrated within like-minded networks, users may face narrower exposure to diverse viewpoints, creating conditions consistent with echo chambers and filter bubbles (Flaxman et al. 2016, Kitchens et al. 2020). In contrast, exposure to heterogeneous audiences increases informational variety, including cross-cutting viewpoints (Bakshy et al. 2015, Barberá et al. 2015). Because collective attention naturally decays over time, continued novelty is important for improving engagement (Wu and Huberman 2007). In this sense, "outsiders" can broaden the informational environment and introduce fresh viewpoints that counteract attention decay.

We argue that SELs can increase engagement by lowering cognitive barriers that otherwise discourage diversified engagement. Theoretically, we posit a mechanism linking cognitive load reduction (a latent constraint) to a behavioral outcome (diversification of engagement). In standard social media environments, users often conserve cognitive resources by focusing engagement on familiar peers and well-known topics rather than expending the high search and processing effort required to venture beyond them (Fiske and Taylor 1991, Pirolli and Card 1999, Stanovich 2009). We propose that SELs act as an overarching informational subsidy that lowers the marginal cost of exploration. Because this cognitive load is reduced, users rationally engage in more exploration, manifesting empirically as diversification along two distinct dimensions.

First, SELs facilitate topic diversification in user participation. In standard social media environments, users often remain within topical silos, concentrating on familiar topics. This is because entering a new domain requires cognitive effort to acquire context and interpret unfamiliar information (Fiske and Taylor 1991, Stanovich 2009). By synthesizing dispersed content into accessible summaries and surfacing clear entry points, SELs reduce the cognitive load of sensemaking and lower the effort required to explore unfamiliar topics (Sweller 1988, Jiang and Benbasat 2007). Because discovery frictions shape concentration, lowering these costs should disproportionately increase engagement with niche topics and expand participation in the long-tail (Brynjolfsson et al. 2011). As a result, users may allocate attention across a broader range of topics, consistent with competitive attention dynamics in online environments (Wu and Huberman 2007, Weng et al. 2012).

Second, SELs may foster participant diversity by mitigating the cognitive barriers associated with cross-network engagement. Breaking out of one's established social graph to interact with unfamiliar users imposes a twofold cognitive burden: the initial search cost required to locate relevant external conversations, and the interpersonal effort needed to decipher unstated norms and evaluate the intent and credibility of strangers (Bakshy et al. 2012, Shi et al. 2014, Goel et al. 2016). We posit that SELs act as informational subsidies that absorb these costs. By actively synthesizing overarching narratives and providing interaction-ready entry points, SELs dramatically lower the friction and perceived risk of cross-cutting exposure. Because the cognitive penalty for interacting with "outsiders" is neutralized, users can afford to engage with a more heterogeneous mix of participants, thereby broadening the participant base of a reply thread. Importantly, this mechanism differs from topic diversity: even when users remain within familiar topics, exposure can extend beyond their existing networks and introduce more cross-cutting perspectives than would arise from friend-centric exposure alone (Bakshy et al. 2015).

Hence, both increased participant diversity and increased topic diversity represent plausible pathways through which SELs enhance engagement. Together, this dual diversification may help sustain engagement by bringing in a broader set of participants and topics. Therefore, we propose:

Hypothesis 3 (H3). *The positive relationship between SEL adoption and user engagement is mediated by increases in participant diversity and topic diversity.*

3 Data and Context

3.1 Research Context and Data Collection

On November 10, 2024, X.com allowed all users in New Zealand to use Grok for free, a feature previously restricted to premium users (Mehta 2024). This release creates a natural experiment that allows us to estimate the causal effect of Grok adoption on user engagement.

We define the treatment group as non-premium users in New Zealand, as this cohort experienced the direct shift from no access to free access on November 10, 2024. Users in Australia serve as the control group, providing a robust counterfactual for New Zealand due to the profound and extensive sociodemographic and institutional similarities between the two nations. As a few examples, both countries operate as parliamentary democracies within the Commonwealth and share foundations in the common-law legal tradition. They also exhibit strong socio-cultural alignment, as evidenced by a shared primary language, highly overlapping media ecosystems, and common cultural touchstones, such as a national passion for sports like rugby and cricket. Crucially for this study, both demonstrate similar levels of technological adoption, with comparable rates of internet and social media usage that suggest a strong baseline similarity in their online behaviors (internet penetration is nearly identical: 94.9% in Australia vs. 95.7% in New Zealand, as is social media usage (78.3% vs. 78.7%) according to DataReportal (2024a, 2024b). These similarities help reduce confounding factors that could bias comparisons.

We collected publicly available data from the X.com platform for users in both countries,[6] covering the period from April 1 to December 6, 2024. We chose December 6 as the end of the estimation period because Grok also became free for all users after this date (Roth 2024). We first identified user locations by leveraging the geotag information embedded in their posts. Location data from user profiles was intentionally avoided because such information is self-reported and may be inaccurate (e.g., users can customize their profile location without verification) (Hecht et al. 2011). Instead, we rely on explicitly geolocated data to map users to geographic boundaries, following recent empirical applications (Ananthakrishnan et al. 2025). This approach provides the spatial precision necessary to separate our treatment and control environments. To construct the sample, the treatment group (New Zealand) included users posting with a geotag within a 100-mile radius of Auckland, the country's most populous area. For the control group (Australia), we applied the same 100-mile radius around Sydney. Given Australia's vast terrain, we chose Sydney as the largest city to ensure the control group is drawn from a comparable metropolitan environment.[7] Importantly, this geographic separation effectively mitigates potential network spillover. Recent geospatial analyses also confirm that platform interactions exhibit strong spatial homophily, keeping cross-border conversational overlap structurally sparse (Wang et al. 2024). Empirical inspection of our dataset confirms this: throughout the observation period, there were zero instances of cross-engagement between the 360 control users and the 510 treated users, ensuring the control baseline remains completely untainted. Further, because Grok functions as a private, user-initiated discovery tool rather than a platform-wide algorithmic shift, Australian users remain structurally isolated from the treatment. Lastly, to avoid contamination from pre-treatment exposure, we excluded Premium subscribers who could access Grok before the rollout to free users[8].

To construct comparable treatment and control groups, we apply Propensity Score Matching (PSM) using seven months of pre-treatment data. This procedure is critical given our sampling method. While Auckland and Sydney serve as strong macro-level counterfactuals, matching users on key pre-treatment covariates further reduces potential confounding and improves comparability between the two groups. Specifically, we conduct matching at the user level via logistic regression, with covariates including the average number of posts, reposts, replies, favorites, and average mobile access during the pre-treatment period, as well

[6] There are two ways we identify the location of users on X.com. In the first approach, we select users whose shared location to be New Zealand or Australia. In the second approach, because users can add a precise location to their post, we include users who generated at least one post in New Zealand or Australia during our estimation period.

[7] While this method ensures geographic precision, we acknowledge the data limitation that our sample only includes the small fraction of users who enable location sharing on their posts (Hecht and Stephens 2014, Pavalanathan and Eisenstein 2015).

[8] According to an estimate by TechCrunch, X.com Premium and Premium+ subscriptions were required for Grok access before the treatment date. They estimate that there were 1.4 million subscribed users on X.com in September 2024 (Perez 2024). Compared with estimated 335 million monthly active users on X.com in 2024 (Statista 2023), roughly 0.4% of X.com users have access to Grok before treatment date.

as the number of followers, friends, total posts, and total favorites of the authors. We perform 1:1 nearest-neighbor matching[9] on the logit of the propensity score, resulting in 269 paired users with strong baseline equivalence. Table 1 demonstrates that PSM successfully balanced these covariates between the treatment and control groups.

Notably, because platform usage logs are unavailable, we cannot verify which users in the treatment group actively engaged with Grok. Our DID estimates, therefore, reflect an intent-to-treat (ITT) effect (i.e., the average effect of having access to Grok) regardless of whether that access was exercised. ITT estimates can be interpreted as a conservative lower bound on the average treatment effect on the treated (ATT).

Table 1. Differences in Mean Before and After Matching

Variables	Before Matching			After Matching		
	Mean Treat	Mean Control	P value	Mean Treat	Mean Control	P value
Log(pre.favorites)	1.720	2.286	0.000	1.903	1.928	0.775
Log(pre.replies)	1.030	1.150	0.001	1.050	1.061	0.782
Log(pre.reposts)	1.248	1.528	0.000	1.341	1.322	0.778
Log(pre.total.posts)	6.147	5.556	0.000	5.984	5.978	0.950
mean.mobile.access	0.573	0.740	0.000	0.689	0.672	0.614
Log(author.followers)	6.814	7.220	0.005	6.773	6.919	0.387
Log(author.friends)	6.339	6.510	0.131	6.433	6.483	0.709
Log(author.total.posts)	9.605	9.050	0.000	9.350	9.371	0.873
Log(author.tot.favorites)	1.268	1.970	0.000	1.503	1.538	0.715

Note: mean.mobile.access = average of mobile_sources (1 = mobile access via iPhone, iPad or Android; 0 = web access) during the pre-treatment period.

The estimation panel dataset is structured at the user-week level, spanning a four-week window before and after Grok's release (October 13 to December 6, 2024). Our primary dependent variables capture the overall volume of audience response to a user's content using engagement metrics (favorites, replies, and reposts). To examine the plausible mechanisms proposed in H3, we further quantify the structural diversity of platform interactions using Shannon entropy. This approach aligns with recent studies measuring diversity in user-engaged content (Zhang et al. 2025), digital behaviors (Wang and Li 2025), and the lexical diversity of user-generated content (Liu et al. 2024). Specifically, we define two entropy-based metrics. First, participant diversity captures the broadening of the audience base, calculated based on the distribution of unique users who authored the replies received by a focal user. Second, topic diversity measures the semantic breadth of a user's own participation, calculated based on the thematic distribution of the user's "@" reply-style activities. The detailed mathematical formulations and aggregation procedures for constructing both diversity metrics are detailed in Appendix D. Table 2 summarizes the key variables used in the analysis.

[9] Our results are consistent with Covariate Balancing Propensity Scores (CBPS) method, as shown in Appendix C.

Table 2. Summary Statistics of Key Variables

Variables	Description	Count	Mean	Stdev	Min	Max
favorites	The average number of favorites received per user-initiated activity by user i during week t.	3,115	12.05	80.51	0	2835.8
replies	The average number of replies received per user-initiated activity by user i during week t.	3,115	0.86	2.35	0	67.3
reposts	The average number of reposts received per user-initiated activity by user i during week t.	3,115	1.79	12.18	0	339.0
replier_entropy	The average Shannon entropy of the participant distribution among users who replied to user-initiated activity of user i during week *t*.	2,417	0.33	0.34	0	2.81
topic_entropy	The average Shannon entropy of the thematic distribution of outgoing "@" reply-style activities made by user i during week t.	2,258	1.58	1.09	0	4.42

Note: replier_entropy measures the social diversity of the audience base engaging with the user's content, while topic_entropy measures the semantic breadth of the user's own participation across different topics. The reduced observation counts for replier_entropy (2,417) and topic_entropy (2,258) reflect weeks in which certain users did not receive any replies or did not author any "@" reply-style activities, respectively.

4 Empirical Model and Main Results

4.1 Empirical Model

To estimate the causal impact of adopting Grok on user behavior, we implement an intent-to-treat (ITT) estimation using a difference-in-differences (DID) regression framework[10]. The main model focuses on engagement with original posts. We take the log of the average number of favorites, replies, and reposts per post at the user-week level, as these variables are highly skewed. The specification is as follows:

$$Engagement_{i,t} = \beta(Treat_i \times After_t) + \eta_i + \theta_t + \varepsilon_{i,t} \tag{1}$$

Here, $Engagement_{i,t}$ represents the three outcome variables at the user-week level. The main explanatory variable is an interaction term, $Treat_i \times After_t$, which equals 1 for users in the treatment group (New Zealand) during the post-rollout period and 0 otherwise. η_i and θ_t denote user and week fixed effects, capturing unobserved heterogeneity across individuals and time. $\varepsilon_{i,t}$ is the error term.

[10] Because platform usage logs are unavailable, treatment is defined as access to Grok rather than confirmed usage. This approach is consistent with prior IS and Gen-AI studies that similarly lack individual-level usage data and estimate ITT effects as a result (Cao et al. 2024, Burtch et al. 2024, Shan and Qiu 2025). ITT estimates the average effect of Grok availability across all treated users and therefore represents a conservative lower bound on the true average treatment effect on the treated (ATT). To the extent that a meaningful fraction of treated users did not engage with Grok, our estimates understate the effect among actual users, lending additional credibility to the positive effects we report.

4.2 Main Results

Table 3. Effect of Grok on Engagement

DV	*Log(favorites)*	*Log(replies)*	*Log(reposts)*
Treat × *After*	0.224*** (0.049)	0.067*** (0.020)	0.084*** (0.030)
User, Period FE	Yes	Yes	Yes
Observations	3,115	3,115	3,115
Adjusted R^2	0.746	0.695	0.764

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01; Weeks in which users have no activities are excluded in the estimation.*

Table 3 presents the estimated treatment effects of Grok adoption on user engagement metrics from Equation (1). The results indicate that the introduction of Grok significantly increases engagement with original posts, with positive effects on favorites ($\beta = 0.224$, $p < 0.001$), replies ($\beta = 0.067$, $p < 0.001$), and reposts ($\beta = 0.084$, $p < 0.01$). These findings support Hypothesis 1 (H1), suggesting that the incorporation of Grok enhances users' engagement with user-generated content on X.com. In this setting, the results are more consistent with facilitation than with substitution.

5 Heterogeneous Effect Analysis

As discussed in Section 2, popular topics already attract considerable attention, leaving less room for Grok to further increase engagement. To empirically test Hypothesis 2, we split the sample into high- and low-popularity topics and estimate treatment effects separately to assess the moderating role of topic popularity on Grok adoption.

5.1 Identifying High- and Low-Popularity Topics

We classify user-generated posts into topics using a semantic clustering pipeline that combines Sentence-BERT (SBERT) and BERTopic. Specifically, we use the pretrained "all-MiniLM-L6-v2" SBERT model (Reimers and Gurevych 2019). This model encodes each post into dense sentence embeddings that capture contextual meaning beyond surface-level keywords. Prior to embedding, all post texts were cleaned by removing user mentions (e.g., @elonmusk), hyperlinks (e.g., URLs or HTML links), and redundant whitespace, while retaining semantically meaningful tokens such as hashtags and normalized emojis. We then apply BERTopic (Grootendorst 2022), which integrates a dimensionality reduction algorithm (UMAP) and a density-based clustering algorithm (HDBSCAN) to group SBERT embeddings into semantically coherent topic clusters. This procedure produces 1,965 unique topics for original posts, which are subsequently mapped back to the dataset for downstream analyses.

To measure topic popularity, we group posts by their topic labels and evaluate each topic's popularity levels during the pre-treatment period (dates prior to November 10, 2024). Specifically, we aggregate engagement by topic over the four weeks immediately pre-treatment (periods -4 to -1, i.e., the one-month window before the intervention). For each topic, we calculate the average of log-transformed engagement across favorites, replies, and reposts within this preceding treatment period.

Prior research shows that attention on digital platforms follows a heavy-tailed distribution, where a small fraction of content captures most user engagement (Susarla et al. 2012, Cinelli et al. 2021). Prior research documents the extreme skew of social media attention, noting that the top 10% of channels often generate nearly 90% of all user interactions (Cinelli et al. 2021). Given this hyper-concentration, we adopt a conservative threshold to isolate viral content. Accordingly, we classify topics as high-popularity only if their pre-treatment averages (periods -4 to -1) exceed the 85th percentile across all metrics, ensuring our definition captures the truly popular topics of the distribution while avoiding borderline cases. Using pre-treatment data also helps mitigate potential endogeneity concerns from treatment-period engagement. Robustness checks using alternative percentile thresholds (80th and 90th) yield consistent results (see Section 7.4 for details). Finally, we construct a user-week panel by aggregating engagement metrics separately for high- and low-popularity topics. For each user in each week, we calculate two distinct sets of engagement totals: one derived from their posts on high-popularity topics and another from their posts on low-popularity topics.

5.2 Heterogeneous Effects by Topic Popularity

We then estimate difference-in-differences (DID) models for high- and low-popularity topic groups separately to assess how topic popularity moderates the impact of Grok adoption.

For high-popularity topics, Grok adoption does not produce statistically significant changes in user engagement (Table 4) across favorites ($\beta = 0.208$, $p > 0.1$), replies ($\beta = 0.025$, $p > 0.1$), and reposts ($\beta = 0.100$, $p > 0.1$). By contrast, for low-popularity topics, Grok produces statistically significant increases in engagement. Treatment effects are positive and significant across all three outcomes: favorites (β=0.236, p<0.001), replies ($\beta = 0.075$, $p < 0.001$), and reposts ($\beta = 0.102$, $p < 0.001$). These results support Hypothesis 2 (H2), demonstrating that the impact of AI assistance is context dependent. While its effect on popular topics is negligible, it substantially boosts engagement in less popular ones. This suggests that Grok may help mitigate the attention inequality that often marginalizes long-tail content on social media platforms. This result validates cognitive load reduction as a mechanism through which Grok increases engagement.

Table 4. Heterogeneous Effect of Grok on Engagement

Subgroup	High-popularity topics			Low-popularity topics		
DV	*Log (favorites)*	*Log (replies)*	*Log (reposts)*	*Log (favorites)*	*Log (replies)*	*Log (reposts)*
Treat × After	0.208 (0.194)	0.025 (0.120)	0.100 (0.126)	0.236*** (0.048)	0.075*** (0.021)	0.102*** (0.030)
User, Period FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	808	808	808	3,104	3,104	3,104
Adjusted R^2	0.697	0.562	0.737	0.737	0.682	0.737

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

6 Mechanism Exploration

Having established that the positive impact of Grok is uniquely concentrated within low-popularity topics (supporting H2), we restrict our mechanism exploration to this subsample. The absence of a significant effect on high-popularity topics reinforces our earlier theoretical argument that SELs offer less benefit in environments where cognitive load is already low. Given this null result, decomposing the high-popularity subsample yields little insight. We therefore focus our mechanism analysis on low-popularity topics, where the treatment effect is concentrated, to examine the pathways through which Grok influences user engagement.

Consistent with our framework, we decompose this impact into two dimensions of diversification: (1) the broadening of the participant base engaging with a post (participant diversity), and (2) the expansion of the semantic breadth of a user's own participation across different topics (topic diversity). We analyze these dimensions in separate subsections for two reasons. First, they represent distinct theoretical mechanisms (as discussed in Section 2.4). While participant diversity captures the broadening of the audience engaging with a focal user's content, topic diversity reflects the user's own behavioral shift toward exploring new conversational themes. Second, the empirical construction of these metrics utilizes different datasets (Appendix D): participant diversity is derived from replies to a user's original posts, while topic diversity is measured via a user's own replies to others. The following sections detail each mechanism in turn.

6.1 Mediation via Participant Diversity

To examine the mechanism through which Grok adoption influences engagement, we employ a causal mediation analysis framework. Mediation analysis decomposes a treatment effect into two components: an indirect effect operating through an intermediate variable (mediator) and a direct effect not transmitted through that mediator (Imai et al. 2010, Tingley et al. 2014). In our context, we first posit that participant diversity partially mediates the relationship between Grok adoption and user engagement. Participant diversity, measured by replier entropy, captures the breadth of the audience-based engagement received in response to a user's activities.

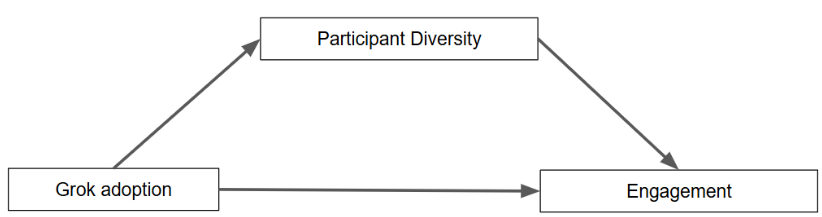

Figure 2. Conceptual Model for Mediation Analysis via Participant Diversity

Following procedures in the causal mediation literature, we implement a two-step estimation approach using the *mediation* package in R (Imai et al. 2010, Tingley et al. 2014, Al Balawi et al. 2023). The conceptual structure of the mediation process is depicted in Figure 2, and the associated models are specified as follows:

In the first stage, we model the mediator (participant diversity) as a function of treatment (Grok adoption), controlling for user and week fixed effects. In the second stage, we model engagement outcomes (favorites, replies, and reposts) as functions of both the treatment and the mediator, with the same fixed effects. Similar to Equation (1), the focal term is the interaction term $Treat_i \times After_t$.

$$Participant\ Diversity_{i,t} = \beta_1 Treat_i \times After_t + \eta_i + \theta_t + \varepsilon_{i,t} \tag{2-1}$$

$$Engagement_{i,t} = \beta_2 Treat_i \times After_t + \delta\ Participant\ Diversity_{i,t} + \eta_i + \theta_t + \varepsilon_{i,t} \tag{2-2}$$

Here, $Engagement_{i,t}$ denotes the log of favorites, replies, or reposts; $Participant\ Diversity_{i,t}$ is the mediator, and η_i and θ_t represent user and week fixed effects. Participant diversity (replier entropy) reflects the distribution of unique users who reply to user *i*. Higher values indicate that replies come from a broader and more diverse set of participants.

As a preliminary step, we examine whether Grok adoption increases participant diversity (Equation 2-1). As shown in Table 5, we find evidence of increased participant diversity following Grok adoption. The treatment effect is positive and statistically significant for replier entropy ($\beta = 0.033$, $p < 0.05$). These results suggest that Grok broadens the pool of participants engaging with posts in the low-popularity subset.

Table 5. Effect of Grok on Participant Diversity (Low popularity topics)

DV	*Log(replier_entropy)*
Treat × After	0.033** (0.015)
User, Period FE	Yes
Observations	2,399
Adjusted R^2	0.488

*Note: Log-transformation was applied to replier entropy due to its high skewness (skewness = 4.60); Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

We then estimate the Average Causal Mediation Effect (ACME) and Average Direct Effect (ADE) using 1,000 quasi-Bayesian Monte Carlo simulations to test the mediating roles of participant diversity. As shown in Table 6, the indirect effect through participant diversity is positive and significant across all engagement measures: favorites (ACME = 0.041, $p < 0.05$), replies (ACME = 0.025, $p < 0.05$), and reposts (ACME = 0.019, $p < 0.05$). These results indicate that part of Grok's effect on engagement is mediated by increased diversity among participants. The direct effects remain positive and significant, suggesting that while participant diversity accounts for a meaningful portion of the overall effect (mediating 15.0% to 32.6% of the total effect), additional mechanisms also contribute to the observed increase in engagement.

Table 6. Causal Mediation Effects via Participant Diversity

DV	*Log (favorites)*		*Log (replies)*		*Log (reposts)*	
	Estimate	95% CI	Estimate	95% CI	Estimate	95% CI
ACME	0.041**	[0.005, 0.076]	0.025**	[0.002, 0.048]	0.019**	[0.002, 0.036]
ADE	0.219***	[0.121, 0.317]	0.051***	[0.015, 0.090]	0.108***	[0.048, 0.171]
Total Effect	0.260***	[0.153, 0.368]	0.076***	[0.031, 0.121]	0.127***	[0.062, 0.191]
Prop. Mediated	0.159**	[0.022, 0.315]	0.326**	[0.049, 0.676]	0.150**	[0.015, 0.323]

*Note: Mediator is Log(replier_entropy); Simulations: 1,000; *: p<0.1, **: p<0.05, ***: p<0.01*

6.2 Mediation via Topic Diversity

We next explore topic diversity as another mechanism. Topic diversity, measured by topic entropy, reflects the semantic breadth of a user's own reply activities. As detailed in Section 3.1 and Appendix D.2, topic entropy is calculated exclusively from "@" reply-style activities. This focus allows us to examine how Grok relates users' engagement with external content beyond their typical interest topics. Higher values suggest that the user is engaging with a more diverse set of topical clusters rather than focusing on a single niche.

It is important to note that the outcome variable in this mediation analysis differs from that in Section 6.1. Because topic diversity is measured via a user's own outgoing reply activities, the second stage of this analysis models the engagement received by those user-initiated replies – that is, the favorites, replies, and reposts that the user's own reply posts attract – rather than engagement on the user's original focal posts. This distinction reflects a difference in the locus of engagement: while Section 6.1 examines whether a more diverse audience engages with a user's posts, Section 6.2 examines whether a user's active participation across diverse topics draws engagement back from the communities they enter. Following the same two-step estimation approach, the conceptual structure is depicted in Figure 3, and the associated models are specified as follows:

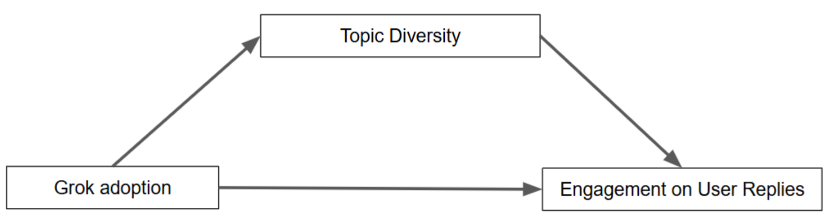

Figure 3. Conceptual Model for Mediation Analysis via Topic Diversity

$$Topic\ Diversity_{i,t} = \beta_1 Treat_i \times After_t + \eta_i + \theta_t + \varepsilon_{i,t} \quad (3\text{-}1)$$

$$Outgoing\ Engagement_{i,t} = \beta_2 Treat_i \times After_t + \delta Topic\ Diversity_{i,t} + \eta_i + \theta_t + \varepsilon_{i,t} \quad (3\text{-}2)$$

As a preliminary step, we find that Grok adoption significantly increases topic diversity ($\beta = 0.139$, $p < 0.05$), as shown in Table 7. This suggests that by lowering informational and cognitive barriers, Grok enables users to navigate and contribute to a broader spectrum of topical clusters, which in turn stimulates a higher frequency of outgoing interactions and dialogue. In other words, Grok expands the topical scope of the user's own contributions to the platform's conversation network.

Following the causal mediation procedures (Table 8), we find that topic diversity is associated with a statistically significant indirect effect for the favorites (ACME = 0.007, $p < 0.1$) and replies (ACME = 0.004, $p < 0.01$) that the user's replies receive. The proportion of the total effect mediated by the topic diversity metric ranges from 2.3% to 4.6%. These findings indicate that navigating a broader spectrum of topics is associated with more diverse reactions and reciprocal dialogue.

Interestingly, the ACME for reposts via topic diversity is not statistically significant (ACME=0.001, $p > 0.1$). This null effect highlights a critical distinction between localized interaction and network-wide broadcasting. SELs successfully lower the cognitive barriers that prevent a user from entering new domains and conversing with unfamiliar peers. However, reposting acts as a structural endorsement that curates what appears before a user's own follower network. The cognitive load reduction provided by Grok thus makes it affordable for users to consume or directly reply to highly diverse, cross-cutting content, but the perceived social risk of broadcasting that off-brand content to their established, homophilous audience remains much higher.

Table 7. Effect of Grok on Topic Diversity (Low popularity topics)

DV	*topic_entropy*
Treat × After	0.139** (0.068)
User, Period FE	Yes
Observations	2,256
Adjusted R^2	0.759

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

Table 8. Causal Mediation Effects via Topic Diversity

DV	*Log (favorites)*		*Log (replies)*		*Log (reposts)*	
	Estimate	95% CI	Estimate	95% CI	Estimate	95% CI
ACME	0.007*	[-0.000, 0.017]	0.004***	[0.001, 0.009]	0.001	[-0.004, 0.005]
ADE	0.268***	[0.163, 0.373]	0.088***	[0.046, 0.129]	0.129***	[0.066, 0.185]
Total Effect	0.275***	[0.169, 0.378]	0.092***	[0.051, 0.134]	0.129***	[0.066, 0.187]
Prop. Mediated	0.023*	[-0.000, 0.066]	0.046***	[0.010, 0.128]	0.003	[-0.034, 0.045]

*Note: Mediator is topic_entropy; Simulations: 1,000; *: p<0.1, **: p<0.05, ***: p<0.01*

Overall, these findings broadly support Hypothesis 3 (H3). Both participant diversity and topic diversity emerge as significant mediating channels, consistent with the dual diversification mechanism proposed in Section 2.4.

7 Robustness Checks and Additional Analysis

In this section, we conduct a series of robustness checks to assess whether our findings are sensitive to modeling choices or data structure. First, we estimate a relative-time (event-study) model to validate the parallel-trends assumption and examine the temporal dynamics of treatment effects (Section 7.1). Second, we replicate the main analyses at the topic-week level to confirm that the results are held under an alternative unit of aggregation (Section 7.2). Third, we exclude topics unique to either treatment or control group (Section 7.3). By restricting the analysis to shared content, we eliminate potential confounding driven by country-specific news cycles or local idiosyncratic events. Fourth, we test the sensitivity of our heterogeneous results to alternative thresholds for defining topic popularity (Section 7.4). Furthermore, we provide additional validation in the Appendix, including topic-level replications with varied popularity thresholds (Appendix A), alternative model specifications with controls (Appendix B), Covariate Balancing Propensity Scores (CBPS) for robust weighting (Appendix C), and counterfactual estimations with placebo tests (Appendix E), all of which consistently support our main conclusions.

7.1 Relative Time Model

To further validate the parallel trends assumption behind the DID model, and assess the temporal dynamics of the treatment effects, we estimate a relative time (event study) model following previous literature (Greenwood and Wattal 2017). The event study specification includes a series of lead and lag indicators relative to the treatment date, allowing us to trace the dynamics of treatment effects before and after Grok's introduction. Specifically, we construct relative time dummies for periods -4 to -1 (pre-treatment weeks) and 0 to +4 (post-treatment weeks), where period -1 serves as the baseline. The model is defined as:

$$Engagement_{i,t} = \sum_{k \neq -1} \beta_k D_{i,t+k} + \eta_i + \theta_t + \varepsilon_{i,t}, \tag{4}$$

where $Engagement_{i,t}$ denotes the engagement outcome (favorites, replies and reposts) for user i in week t, $D_{i,t+k}$ is an indicator for being in relative week k with respect to the treatment date (excluding -1 as the omitted category), and η_i are user fixed effects, θ_t are period fixed effects.

Figure 4 presents the estimated coefficients from the event-study model. For the full sample (left panel), the coefficients on the lead terms are statistically insignificant across all engagement metrics, supporting the validity of the parallel-trends assumption. The post-treatment coefficients are generally positive and consistent with the main DID results, indicating an increase in engagement following Grok's introduction. The 95% confidence intervals (shown by the shaded bands) illustrate that the effect persists beyond the immediate post-treatment period. These findings provide evidence that the treatment effects reflect sustained behavioral changes.

We further replicate the relative-time model for the low-popularity topic subset, where Grok's impact is most pronounced. As shown in Figure 4 (right panel), the pre-treatment coefficients are insignificant, while the post-treatment coefficients are positive and significant across favorites, replies, and reposts. These results confirm that the parallel trend assumption also holds for low-popularity topics.

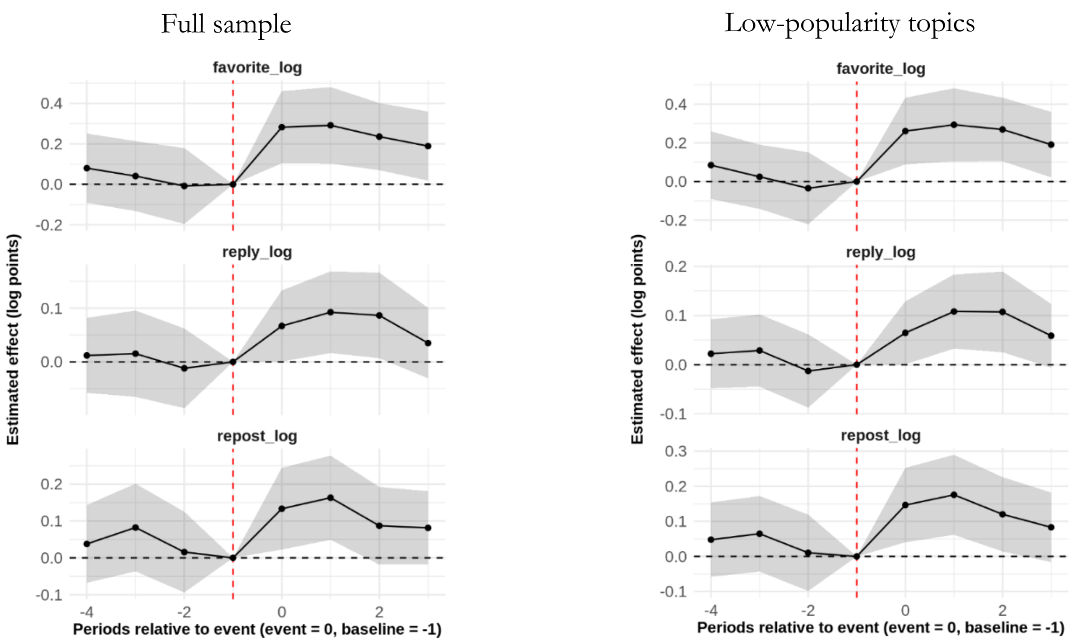

Figure 4. Relative Time Estimates of Engagement

7.2 Topic Level Analysis

While our previous heterogeneous analyses support H2, those models are estimated at the user-week level. To further assess the robustness of these findings, we conduct additional heterogeneous analyses at the topic-week level. This approach allows us to compare treatment effects across topics rather than individual users, providing an alternative robustness test on our estimated impact of Grok's introduction on engagement dynamics across different content topics. Specifically, we identify discussion topics using a classification procedure with SBERT and BERTopic (as described in Section 5) and extract 1,965 unique topics from users' posts. Based on these identified topics, we construct a new panel dataset by aggregating all engagement activities at the topic-week level. Each observation thus represents the aggregate engagement metrics (e.g., favorites, replies, reposts) for posts associated with a given topic in each week.

In this section, we perform two complementary analyses to validate and extend our main findings. Section 7.2.1 tests whether the overall treatment effects remain robust when engagement metrics are aggregated at the topic–week level. Section 7.2.2 explores whether our estimated results for H2 are robust in this robustness analysis. This approach provides a more direct comparison of treatment effects between high- and low-popularity topics.

7.2.1 Topic Level Effects: Overall Results

We first use the topic-level analyses to validate the robustness of our main findings on the overall effect of Grok adoption. Table 9 presents the treatment effects when engagement metrics are aggregated at the topic level. The results are consistent with the user level analysis: Grok adoption leads to significant increases in favorites, replies and reposts. These findings confirm that our main results are robust and not sensitive to the unit of analysis (i.e., user-level vs. topic-level).

Table 9. Effects of Grok on Engagement (Topic Level Analysis)

DV	*Log(favorites)*	*Log(replies)*	*Log(reposts)*
Treat × After	0.184*** (0.031)	0.035** (0.015)	0.056*** (0.015)
Topic, Period FE	Yes	Yes	Yes
Observations	16,498	16,498	16,498
Adjusted R^2	0.421	0.363	0.485

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

7.2.2 Heterogeneous Effects by Topic Popularity (Topic Level Analysis)

Similar to our analyses in Section 5, we use the 85th percentile as the cutoff point between high and low popularity topics. We next conduct the heterogeneous analysis at the topic level and present the results in Table 10. As presented in Table 10, the results are consistent with our main analyses at the user-week level. For low-popularity topics, treatment effects remain positive and statistically significant across all engagement measures (columns 4-6 in Table 10). By contrast, for the high-popularity topic group, treatment effects are not statistically significant (columns 1-3 in Table 10). This finding shows that while Grok adoption yielded no significant gains for already-prominent topics, it meaningfully enhances engagement for content that otherwise struggles to gain organic attention.

Table 10. Heterogeneous Effects of Grok on Engagement (Topic Level Analysis)

Subgroup	*High-popularity topics*			*Low-popularity topics*		
DV	*Log (favorites)*	*Log (replies)*	*Log (reposts)*	*Log (favorites)*	*Log (replies)*	*Log (reposts)*
Treat × After	0.177 (0.274)	0.073 (0.145)	0.071 (0.187)	0.184*** (0.030)	0.034** (0.014)	0.055*** (0.014)
Topic, Period FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	597	597	597	15,774	15,774	15,774
Adjusted R^2	0.501	0.390	0.580	0.376	0.329	0.385

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

7.3. Analysis Excluding Country-Specific Topics

Our main identification strategy relies on propensity score matching (PSM) to mitigate potential selection bias. However, if certain topics receive disproportionate attention in only one country at a specific time (e.g., "*newzealand*", "*australia*", "*earthquake*", "*maori*"), one might question whether the treatment and control groups are truly comparable. In other words, topic differences between the treatment group and the control group could bias the estimates. To address this issue regarding topic confounding effects, we conduct a robustness check by removing topics that appear exclusively in either the treatment or control group. We use SBERT-based topic classification to identify topics that are present only in one country. Posts associated with these topics are excluded, and the DID models are re-estimated using the filtered dataset.

7.3.1 Effects after Removing Group-Specific Topics: Overall Results

Table 11 presents treatment effects on user engagement after removing group-specific topic keywords. The results remain consistent with the main findings: Grok adoption continues to significantly increase favorites, replies and reposts.

Table 11. Effects of Grok on Engagement (Group-Specific Topics Removed)

DV	*Log(favorites)*	*Log(replies)*	*Log(reposts)*
Treat × After	0.175*** (0.049)	0.051*** (0.019)	0.052* (0.031)
User, Period FE	Yes	Yes	Yes
Observations	3,074	3,074	3,074
Adjusted R^2	0.735	0.682	0.746

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

7.3.2 Heterogeneous Effects by Topic Popularity (after Removing Group-Specific Topics)

One may argue that our heterogeneous treatment effect results in Section 5 could be confounded by country-specific conversations that only users in one of the groups are interested in. Thus, we repeat the analyses in Section 5 but excluding country-specific topics. For low-popularity topics, the treatment effects remain positive and statistically significant across all outcomes (Table 12). These findings reinforce that the adoption of Grok meaningfully enhances engagement in contexts where posts receive less organic attention. For high-popularity topics, the treatment effects remain statistically insignificant. These results align with the previous findings. This confirms that the main findings are not driven by topics unique to one group.

Table 12. Heterogeneous Effects of Grok (Group Specific-Topics Removed)

Subgroup	High-popularity topics			Low-popularity topics		
DV	*Log (favorites)*	*Log (replies)*	*Log (reposts)*	*Log (favorites)*	*Log (replies)*	*Log (reposts)*
Treat × After	-0.056 (0.224)	-0.042 (0.131)	0.059 (0.151)	0.193*** (0.049)	0.057** (0.019)	0.073** (0.030)
User, Period FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	638	638	638	3,065	3,065	3,065
Adjusted R^2	0.727	0.606	0.764	0.724	0.663	0.719

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

7.4 Analysis with Alternative Thresholds for Topic Popularity

To assess whether our findings are sensitive to the definition of high-popularity topics, we conduct a robustness check using alternative thresholds for classifying topic popularity. In the main analysis, high-popularity topics are defined as those exceeding the 85th percentile of favorites, replies, and reposts during the pre-treatment period. Although this cutoff effectively captures consistently popular topics, we acknowledge that threshold selection can influence categorization. To ensure our findings are not artifacts of this specific choice, we conduct sensitivity analyses using alternative cutoffs (e.g., 80th and 90th percentiles). This approach allows us to examine whether effects vary across different definitions of topic popularity, thereby reinforcing the robustness of our heterogeneous analysis framework.

Table 13 reports the treatment effects on engagement for topics below the 80th and 90th percentile thresholds. Across both specifications, the treatment effects remain positive and statistically significant for favorites, replies and reposts.

Table 13. Effects on Engagement for Low-Popularity Topics (Alternative Thresholds)

Subgroup	*< 80th Percentile*			*< 90th Percentile*		
DV	*Log (favorites)*	*Log (replies)*	*Log (reposts)*	*Log (favorites)*	*Log (replies)*	*Log (reposts)*
Treat × After	0.237*** (0.048)	0.076*** (0.021)	0.101*** (0.030)	0.226*** (0.048)	0.071*** (0.020)	0.095*** (0.030)
User, Period FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,100	3,100	3,100	3,107	3,107	3,107
Adjusted R^2	0.730	0.678	0.727	0.739	0.687	0.739

*Note: Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

We further examine whether the results for diversity (participant, topic diversity) are consistent under these alternative thresholds. As shown in Table 14, Grok adoption continues to significantly increase participant diversity, measured by replier entropy. These results indicate that the positive treatment effects on both engagement and participant diversity are robust to alternative definitions of low-popularity topics. Furthermore, to verify that these threshold sensitivities are not driven by the choice of aggregation unit, we replicated this analysis using topic-level aggregation. As detailed in Appendix A, the treatment effects remain highly consistent.

Table 14. Effects on Diversity for Low-Popularity Topics (Alternative Thresholds)

Subgroup	*< 80th Percentile*		*< 90th Percentile*	
DV	*Log(replier_entropy)*	*topic_entropy*	*Log(replier_entropy)*	*topic_entropy*
Treat × After	0.032** (0.015)	0.139** (0.068)	0.030** (0.015)	0.140** (0.067)
User, Period FE	Yes	Yes	Yes	Yes
Observations	2,392	2,256	2,403	2,257
Adjusted R^2	0.485	0.759	0.494	0.759

*Note: Topic entropy is retained in its raw form as its distribution is approximately symmetric (skewness = 0.16), unlike replier entropy which was log-transformed due to high skewness (skewness = 4.60); Robust standard errors in parentheses; *: p<0.1, **: p<0.05, ***: p<0.01*

8. Discussion

The growing literature on Gen-AI has documented substantial individual-level gains from stand-alone LLM tools such as ChatGPT, in productivity (Noy and Zhang 2023), creative performance (Zhou and Lee 2024, Doshi and Hauser 2024), and online knowledge contribution (Shan and Qiu 2025). However, this literature predominantly conceptualizes Gen-AI as an endpoint, where Gen-AI is designed to optimize isolated tasks in situ, with limited attention to how AI might reshape the interdependent dynamics of social interaction. This study reorients this framing by theorizing and empirically validating *Socially Embedded LLMs* (SELs) as a qualitatively distinct class of generative systems. Rather than functioning as passive assistants, they operate as active social intermediaries embedded within the platform. Unlike stand-alone LLMs as information silos, which provide synthesized answers without an easy mechanism to refer users back to the social media source content, SELs embedded within social platforms may alter the economics of interaction. Meanwhile, compared with traditional search, which requires users to manually forage through fragmented results, SELs synthesize dispersed information while providing direct links, thereby substantially reducing the cognitive load required for participation. Our findings from Grok's staged rollout on X.com provide early causal evidence that, when designed with embedded social entry points, an LLM can function as a facilitator of human discourse rather than a substitute for it.

The facilitation result has a clear structural explanation. Grok's design eliminates the friction that sustains the substitution dynamic in stand-alone LLMs: instead of requiring users to navigate from an AI summary to an external source, Grok's embedded citations place them one click from the original post. This converts synthesis from a terminal activity, where the AI answer satisfies the query and engagement ends, into a conduit activity, where the AI answer primes the user for subsequent social interaction. The broader implication is that the substitution-versus-facilitation question is not determined solely by user motivations or cognitive tendencies, but by the specific architectural choices that govern how AI output connects (or fails to connect) users to the community. Platform designers who embed interaction pathways within AI responses may be better positioned to sustain the organic feedback loops that drive network value.

The heterogeneity result (i.e., Grok's engagement effects are concentrated entirely among low-popularity topics, with no significant change for high-popularity topics) reveals the conditions under which SELs add value. The key mechanism is discovery cost. Highly popular topics already benefit from multiple overlapping discovery affordances (e.g., trending modules, hashtags, and algorithmic ranking). Those affordances collectively ensure that relevant discussions are salient and accessible (Huszár et al. 2022, Yang and Peng 2022, Zhang and Ng 2023). In such environments, SELs are redundant at the margin. In contrast, low-popularity or long-tail topics face higher discovery barriers and greater cognitive effort to locate (Pirolli and Card 1999, Chi et al. 2000, Wu and Huberman 2007). In these peripheral discussions, SELs provide the greatest benefit by surfacing overlooked content and offering interaction-ready entry points for participation. Consequently, these agents may help reallocate attention and engagement toward the long-tail of discussion on the platform.

The mediation analysis identifies diversification of both participants and topics as the primary behavioral channel through which the cognitive relief provided by Grok translate into engagement gains. This finding contributes to the echo chamber and filter bubble literature (Bakshy et al. 2015, Flaxman et al. 2016, Kitchens et al. 2020) in a non-obvious way: while algorithmic personalization is typically blamed for reinforcing homophilous exposure, our theoretical lens suggests that it is also fundamentally driven by users conserving cognitive resources in high-friction environments. Our results indicate that SELs can counteract these tendencies by acting as cognitive subsidies. By synthesizing unfamiliar contexts and absorbing the search and interpersonal evaluation costs of cross-network engagement, SELs make it cognitively affordable for users to venture beyond their familiar social proximity and typical interest clusters. The result is a more heterogeneous conversation, driven not by algorithmic, but by the removal of cognitive barriers.

The findings also have implications for platform design and governance. First, social media platforms can reinforce homophily by exposing users primarily to content from users' existing networks, which may contribute to echo chambers and concentrated information exposure (Bakshy et al. 2015, Flaxman et al. 2016, Kitchens et al. 2020). Our results demonstrate that SELs like Grok can help mitigate these effects by surfacing relevant content from across the platform and routing users to discussions beyond their established networks. For platform managers, these results indicate that SELs can be designed not only to enhance engagement but also to facilitate more diverse and cross-network interaction, supporting a richer exchange of perspectives (Huszár et al. 2022, Guess et al. 2023).

Second, our results suggest that SELs need not be deployed uniformly. Their marginal value is greatest in the areas of the platform that are most underserved by existing discovery mechanisms, such as domains like niche communities, emerging topics, and peripheral discussions that struggle to gain organic traction (Merton 1968, Salganik et al. 2006, Weng et al. 2012). Targeted SEL deployment in these areas could help redistribute engagement toward the long-tail, further strengthening the platform's content diversity without relying on interventions that risk disrupting already-healthy popular discussions. Together, and more broadly, our findings suggest a design principle: AI agents intended to sustain social engagement should be engineered to serve as intermediaries through brokering connections, lowering entry costs, and routing users back to the community, rather than as efficient endpoints that satisfy queries in isolation.

Although the natural experiment provides a strong foundation for causal inference, several limitations merit acknowledgement. First, the findings are specific to Grok's implementation on X.com, a platform with distinctive features and interaction norms. Because the design and integration of SELs may vary across platforms, such differences could influence how users interact with them and the resulting effects on engagement. Therefore, future research could assess whether our findings generalize to other social media contexts, contributing to a broader understanding of how SELs shape digital social environments. Nonetheless, we believe the underlying foundation is theoretically generalizable: AI intermediation can reduce the cognitive search costs. While platform interfaces differ, many platforms confront information overload, suggesting that SELs could offer comparable discovery benefits across other large-scale content ecosystems.

Second, our design estimates intent-to-treat effects (the effect of feature availability rather than actual usage). Because not all users with access to Grok engage with the feature, our estimates reflect average exposure that may mask variation in adoption intensity across users. Consequently, these estimates should be interpreted as conservative lower-bound estimates, while acknowledging that the effect on active adopters may be larger than what we observe. Future work leveraging granular usage logs could isolate the effect of adoption intensity to quantify this upper bound more precisely.

Third, our study relies on explicitly geotagged post data to determine user location. While this methodological choice prioritizes the geographic accuracy required for defining our treatment and control groups, we acknowledge that this opt-in behavior may limit the generalizability of the findings to the broader user base. Accordingly, our findings should be interpreted as reflecting the engagement dynamics of this specific location-sharing subset of users. Finally, our analysis focuses on observable behavioral outcomes, such as engagement levels and participant diversity, leaving open questions about users' perceptions, trust, and longer-term adaptation. Future research that integrates behavioral data with surveys, interviews or experiments could offer a more comprehensive understanding of how SELs influence both engagement dynamics and user perceptions of SEL adoption.

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